

Writing the $\frac{1}{\bar{\epsilon}}$ term as a fragmentation function

Which term it is, the two things that must be fixed first, and the LaTeX to paste

$$\alpha_s = \frac{g^2}{4\pi} \Rightarrow \frac{\alpha_s}{2\pi} = \frac{g^2}{8\pi^2}, \quad \tilde{N}(k) \equiv k^+ \frac{d^3 N}{d^3 k}, \quad \mathcal{X} = \frac{(x-y) \cdot (x'-y')}{(x-y)^2 (x'-y')^2}, \quad r = y' - y, \quad \xi_0 = \frac{k^+}{\sqrt{}}$$

The term is $\frac{\mathcal{P}_{gg}(\xi)}{2N_c} \frac{1}{\bar{\epsilon}} \mathcal{X}$ — the $\mathcal{P}_{gg}$ half of the single pole, and nothing else. It does go into an $\overline{\text{MS}}$ fragmentation function, and it matches the canonical $\frac{\alpha_s}{2\pi} \frac{1}{\bar{\epsilon}} \mathcal{P}_{gg}$ form *exactly* — ratio 1 — but only after two repairs: the convolution has to be written for the **invariant density** $\tilde{N} = k^+ \frac{d^3 N}{d^3 k}$, not for $\frac{d^3 N}{d^3 k}$ itself, and the $\log(4\pi^2 \mu^2 r^2)$ has to stay **inside** the transverse integral. With those two done, this row supplies exactly **half** the counterterm; the mirror contraction supplies the other half. The $\delta(1-\xi) \log \frac{k^+}{\Lambda}$ half of the pole is untouched by any of this.

My reservation, stated once and then set aside. I still read this pole as the large- $|z|$ divergence that cancels against the companion rows rather than a collinear one. You and your advisor want it written as fragmentation, so everything below is that, done properly, with the checks that will tell you if it is right.

1. Isolating the term

From the reduced form of the row, the pole appears once:

$$\frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{N_c}{k^+} \frac{1}{\bar{\epsilon}} \int_{\xi_0}^1 \frac{d\xi}{\xi^2} \left[\underbrace{\delta(1-\xi) \log \frac{k^+}{\Lambda}}_{\text{not fragmentation}} + \underbrace{\frac{\mathcal{P}_{gg}(\xi)}{2N_c}}_{\text{this is the one}} \right] \int_{x,x',y,y'} e^{-i\frac{k}{\xi} \cdot r} \mathcal{X} \mathcal{U}.$$

Only the second entry has the four fragmentation signatures (measure $\frac{d\xi}{\xi^2}$, hard factor at $\frac{k}{\xi}$, splitting function, parent Wilson lines). Take it alone.

2. Repair 1: the convolution needs the invariant density

This is the step that decides whether the factorised form is even writable, so do it explicitly. Suppose your LO row is

$$\frac{d^3 N_{\text{LO}}}{d^3 k}(k) = \frac{1}{(2\pi)^3} \frac{g^2}{\pi^2} \frac{1}{k^+} \int_{x,x',y,y'} e^{-ik \cdot r} \mathcal{X} \mathcal{U}.$$

Evaluated at the *parent* momentum $\frac{k}{\xi}$, **every** component scales — including $k^+ \rightarrow \frac{k^+}{\xi}$ — so

$$\frac{d^3 N_{\text{LO}}}{d^3 k}\left(\frac{k}{\xi}\right) = \frac{1}{(2\pi)^3} \frac{g^2}{\pi^2} \underbrace{\frac{\xi}{k^+}}_{\text{note the } \xi} \int e^{-i\frac{k}{\xi} \cdot r} \mathcal{X} \mathcal{U}.$$

With that, $\frac{\alpha_s}{2\pi} \frac{1}{\bar{\epsilon}} \int \frac{d\xi}{\xi^2} \frac{\mathcal{P}_{gg}}{2} \frac{d^3 N_{\text{LO}}}{d^3 k}\left(\frac{k}{\xi}\right)$ comes out ξ times too big — row/candidate = $\frac{1}{\xi}$ (sympy). The convolution would need $\frac{d\xi}{\xi^3}$, which is not the fragmentation measure. **So the row cannot be written as a fragmentation convolution of $\frac{d^3 N}{d^3 k}$.**

The cure is the standard one: collinear factorisation is stated for the *invariant* single-particle distribution, $E \frac{d\sigma}{d^3 p}$, which in light-cone variables is $\tilde{N}(k) \equiv k^+ \frac{d^3 N}{d^3 k}$ (up to a constant, the rapidity density $\frac{dN}{dy d^2 k_{\perp}}$). It has **no leftover** k^+ :

$$\tilde{N}_{\text{LO}}\left(\frac{k}{\xi}\right) = \frac{k^+}{\xi} \cdot \frac{d^3 N_{\text{LO}}}{d^3 k}\left(\frac{k}{\xi}\right) = \frac{1}{(2\pi)^3} \frac{g^2}{\pi^2} \int e^{-i\frac{k}{\xi} \cdot r} \mathcal{X} \mathcal{U},$$

and now the match is exact (sympy: ratio = 1):

$$\left. \frac{d^3 N^{\text{NLO}}}{d^3 k} \right|_{\frac{1}{\bar{\epsilon}}} = \frac{\alpha_s}{2\pi} \frac{1}{\bar{\epsilon}} \frac{1}{k^+} \int_{\xi_0}^1 \frac{d\xi}{\xi^2} \frac{\mathcal{P}_{gg}(\xi)}{2} \tilde{N}_{\text{LO}}\left(\frac{k}{\xi}\right), \quad \tilde{N} \equiv k^+ \frac{d^3 N}{d^3 k}.$$

3. Repair 2: the log cannot come outside

The same test applied to the *finite* partner: $\log(4\pi^2\mu^2r^2)$ depends on $r = y' - y$, which is an integration variable inside $\tilde{N}_{\text{LO}}$. A convolution coefficient must not. So the pole factorises but its companion logarithm does not; write it out long-hand, with the log next to $\mathcal{X}$ under $\int_{x,x',y,y'}$. This is physics, not bookkeeping: in the CGC the hard scale of the collinear logarithm is the dipole size, and dipole size lives under the transverse integral.

4. The fragmentation function and the subtraction

Define the bare gluon-to-gluon fragmentation function and its $\overline{\text{MS}}$ renormalisation

$$D_{g/g}^{\text{bare}}(\xi) = \delta(1 - \xi) - \frac{\alpha_s}{2\pi} \frac{1}{\bar{\epsilon}} \mathcal{P}_{gg}(\xi) + O(\alpha_s^2), \quad \frac{1}{\bar{\epsilon}} = \frac{1}{\epsilon} - \gamma_E + \log 4\pi,$$

so that the observable $\tilde{N} = \int \frac{d\xi}{\xi^2} D^{\text{bare}} \otimes \tilde{N}^{\text{partonic}}$ is finite, and the renormalised $D_{g/g}(\xi, \mu_F)$ obeys

$$\mu_F^2 \frac{\partial}{\partial \mu_F^2} D_{g/g}(\xi, \mu_F) = \frac{\alpha_s}{2\pi} \int_{\xi}^1 \frac{du}{u} \mathcal{P}_{gg}(u) D_{g/g}\left(\frac{\xi}{u}, \mu_F\right).$$

Operationally: subtract, then rename $\mu \rightarrow \mu_F$.

$$\left. \frac{d^3 N^{\text{NLO}}}{d^3 k} \right|_{\overline{\text{MS}}}(\mu_F) = \left. \frac{d^3 N^{\text{NLO}}}{d^3 k} \right|_{\text{bare}} - \frac{\alpha_s}{2\pi} \frac{1}{\bar{\epsilon}} \frac{1}{k^+} \int_{\xi_0}^1 \frac{d\xi}{\xi^2} \mathcal{P}_{gg}(\xi) \tilde{N}_{\text{LO}}\left(\frac{k}{\xi}\right).$$

Sign check via DGLAP. What survives is $-\frac{\alpha_s}{2\pi} \frac{\mathcal{P}_{gg}}{2} \log(4\pi^2\mu_F^2 r^2)$, whose $\frac{\partial}{\partial \log \mu_F^2}$ is $-\frac{\alpha_s}{2\pi} \frac{\mathcal{P}_{gg}}{2}$; that is cancelled by $\mu_F^2 \partial_{\mu_F^2} D$ acting on the LO term. The μ_F -dependence cancels, as it must.

5. The factor $\frac{1}{2}$, and the prefactor you changed

Comparing the row against the *full* counterterm $\frac{\alpha_s}{2\pi} \frac{1}{\bar{\epsilon}} \mathcal{P}_{gg}$ gives ratio $\frac{1}{2}$ (sympy). That is not an error — it is information:

prefactor on $\frac{\mathcal{P}_{gg}}{2}$	equals	what it is
$\frac{g^2}{8\pi^2}$ (your first version)	$\frac{\alpha_s}{2\pi} \cdot \frac{\mathcal{P}_{gg}}{2}$	what Row IV actually contains
$\frac{g^2}{4\pi^2}$ (your second version)	$\frac{\alpha_s}{2\pi} \cdot \mathcal{P}_{gg}$	the full $\overline{\text{MS}}$ counterterm

Both are correct answers to different questions, which is why the factor 2 looked ambiguous last time. Row IV gives half the counterterm; the mirror contraction (the $y \leftrightarrow y'$ / hermitian-conjugate partner) must give the other half. **So do not subtract the full $\mathcal{P}_{gg}$ against this row alone** — you would over-subtract by a factor 2 and be left with a spurious $-\frac{\mathcal{P}_{gg}}{2\bar{\epsilon}}$. Either subtract $\frac{\mathcal{P}_{gg}}{2}$ here, or wait until the mirror row is in hand and subtract $\mathcal{P}_{gg}$ from the pair. Finding that mirror row and confirming it doubles this one is the single best check that the fragmentation reading is right.

6. What the $\delta(1 - \xi)$ half does

Nothing here touches it. The counterterm is $\frac{\alpha_s}{2\pi} \frac{1}{\bar{\epsilon}} \mathcal{P}_{gg}(\xi)$ — a function of ξ with no $\log \frac{k^+}{\Lambda}$ anywhere in it, because a fragmentation function is defined at a factorisation scale, not at a rapidity cutoff. So after the subtraction you are left with

$$\frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{N_c}{k^+} \left[\log \frac{\vee}{\Lambda} + \log \frac{k^+}{\vee} \right] \frac{1}{\bar{\epsilon}} \int_{x,x',y,y'} e^{-ik \cdot r} \mathcal{X} \mathcal{U}$$

still carrying a pole. Write it down explicitly as the outstanding item rather than letting it disappear — it is the thing that has to cancel against the companion Group II rows, and stating it plainly is what turns the “ $1/\epsilon$?! ” objection into a concrete next calculation.

7. The LaTeX

$$\begin{align}
& \frac{d^3 N_{\rm NLO}}{d^3 k} \Big|_{\rm frag} \\
& \equiv \frac{\frac{1}{\alpha_s} \{2\pi\} \, \overline{\epsilon}}{\int_{\xi_0}^1 \frac{d\xi}{\xi^2} \, \mathcal{P}_{\rm gg}(\xi)} \, \frac{1}{\widetilde{N}_{\rm L0}} \left(\frac{k}{\xi} \right) \\
& \notag \\
& \equiv \frac{\frac{1}{\alpha_s} \{2\pi\} \, \overline{\epsilon}}{\int_{\xi_0}^1 \frac{d\xi}{\xi^2} \, \mathcal{P}_{\rm gg}(\xi)} \, \frac{1}{\frac{(2\pi)^3}{g^2} \pi^2} \\
& \quad \int_{x,x',y,y'} e^{-i \frac{k}{\xi} \cdot (y-y)} \, \frac{1}{\mathcal{F}(x-y) \cdot (x'-y)} \{ (x-y)^2 (x'-y)^2 \} \\
& \notag \\
& \hspace{4.2cm} \times \left[U^{e^+}(y) - U^{e^+}(x') \right] \\
& \quad \left[U^{e^-}(y) - U^{e^-}(x) \right] \rho^{e^+}(x') \rho^{e^-}(x) \, , \\
& \notag \\
& \widetilde{N}(k) \equiv \frac{d^3 N}{d^3 k} \, , \\
& \quad \quad \quad \\
& \alpha_s = \frac{g^2}{4\pi} \, , \\
& \quad \quad \quad \\
& \frac{1}{\overline{\epsilon}} = \frac{1}{\epsilon} - \gamma_E + \log 4\pi \, , \\
& \notag \\
D^{\rm bare}_{\rm g/g}(\xi) &= \delta(1-\xi) \\
& \quad - \frac{1}{\alpha_s} \{2\pi\} \, \overline{\epsilon} \, \mathcal{P}_{\rm gg}(\xi) \\
& \quad + \mathcal{O}(\alpha_s^2) \, , \\
& \notag \\
& \frac{d^3 N_{\rm NLO}}{d^3 k} \Big|_{\overline{\rm MS}} (\mu_F) \\
& \equiv \frac{d^3 N_{\rm NLO}}{d^3 k} \Big|_{\rm bare} \\
& \quad - \frac{1}{\alpha_s} \{2\pi\} \, \overline{\epsilon} \, \frac{1}{k^+} \\
& \quad \int_{\xi_0}^1 \frac{d\xi}{\xi^2} \, \mathcal{P}_{\rm gg}(\xi) \, , \\
& \quad \frac{1}{\widetilde{N}_{\rm L0}} \left(\frac{k}{\xi} \right) \, . \\
& \end{align}$$

Remember when you paste it: the $\frac{\mathcal{P}_{gg}}{2}$ in the first two lines is what *this row* gives; the $\mathcal{P}_{gg}$ in the last line is the *full* counterterm, valid once the mirror row is included. And the finite partner, $-\frac{\alpha_s}{2\pi} \frac{\mathcal{P}_{gg}}{2} \log(4\pi^2 \mu_F^2 r^2)$, must be written with the logarithm inside $\int_{x,x',y,y'}$ — not factored out in front of $\widetilde{N}_{\text{LO}}$.

8. Three checks before you show it to anyone

1. **The ξ power.** Form the ratio $\frac{\text{row's pole term}}{\frac{\alpha_s}{2\pi} \frac{1}{k^+} \int \frac{d\xi}{\xi^2} \frac{\mathcal{P}_{gg}}{2} \widetilde{N}_{\text{LO}}(k/\xi)}$ from *your* LO row. It must be a pure number. If a ξ or $\frac{1}{\xi}$ survives, you have the wrong LO density (§2) or a missing ξ upstream in the $\frac{1}{k^+}$ versus $\frac{1}{S}$ bookkeeping of (0.2).
2. **The *mirror* row.** It should double the $\frac{\mathcal{P}_{gg}}{2}$ to $\mathcal{P}_{gg}$. If it does not, the fragmentation reading is in trouble and the cancellation reading gains.
3. **μ_F -independence.** Differentiate the subtracted result with respect to $\log \mu_F^2$ and check it cancels against $\mu_F^2 \partial_{\mu_F^2} D_{g/g}$. This is the one test a wrong sign or a wrong factor cannot survive.

Checks: `src/rowG2_ff.py` ($\frac{\alpha_s}{2\pi} = \frac{q^2}{8\pi^2}$, the $\frac{1}{\xi}$ failure of the naive density, the exact match with $\tilde{N}$, the factor $\frac{1}{2}$, the DGLAP sign), `src/rowG2_dglap.py` (the $\mathcal{P}_{qq}$ identity and the fragmentation measure), `src/rowG2_sectors3.py` (the three-way split).